L14 Nonhomogeneous Equations: The Method of Undetermined Coefficients

Section 4.4

In this section we find particular solutions y_p to **non-homogeneous** linear equations with constant coefficients

$$ay'' + by' + cy = f(t), \tag{1}$$

where $f(t) \neq 0$.

A **particular solution**, y_p, of equation (1) is a function that contains no arbitrary constants and satisfies equation (1).

ex. Verify that $y_p(t) = \frac{3}{5}\sin t$ is a particular solution of $y'' - 4y = -3\sin t$.

$y = \frac{3}{5}\sin(t)$, $y' = \frac{3}{5}\cos(t)$, $y'' = -\frac{3}{5}\sin(t)$

So $y'' - 4y = \left(-\frac{3}{5}\sin(t)\right) - 4\left(\frac{3}{5}\sin(t)\right)$

$= -\frac{3}{5}\sin(t) - \frac{12}{5}\sin(t) = -\frac{15}{5}\sin(t) = -3\sin(t)$ ✓

$ay'' + by' + cy = f(t)$

Method of undetermined coefficients

right hand side

1. Use the form of f to "guess" a form for a particular solution
2. Plug the guessed form of the particular solution into the DE
3. Determine the coefficients that make your guessed solution an actual equation

$y'' - 4y = -3\sin(t)$

"guess" : $A\sin(t)$

Note that this method is applicable when

1. The DE is linear
2. Coefficients are constants

$ay'' + by' + cy = f(t)$

3. The non-homogeneous term $f(t)$ is of one of the types:

solutions to homogeneous equations

- constants (ex. $f(t) = 10$)
- polynomial (ex. $f(t) = t^2 - 5$, also includes the previous constant case)
- Exponential function $e^{\alpha t}$ (ex. $4t - 1 + e^{3t}$)
- Sine or Cosine functions (ex. $te^t \sin t + (3t-1)e^t$)
- A linear combination of the above (ex. $P(t)e^{\alpha t}\sin\beta t$, where α, β real numbers)

This method is NOT applicable when $f(t) = \ln t, 1/t, \tan t, \cdots$.

ex. Find a particular solution:

1. $y'' + 3y' + 2y = 3t$ — $f(t) = 3t$ polynomial linear

What kind of function y would generate a linear function, $3t$, if we add y and its first and second derivatives?

Try a linear function!

Let $\quad y = At + B$

Then $\quad y' = A \qquad y'' = 0$

Hence $\quad y'' + 3y' + 2y = \underset{y''}{0} + \underset{y'}{3A} + \underset{y}{2(At+B)} \overset{\text{want}}{=} 3t$

Solve systems of 2 equations:

$\Rightarrow \quad 2At + (3A + 2B) = 3t + 0$

$$\Rightarrow \begin{cases} 2A = 3 \longrightarrow A = \frac{3}{2} \\ 3A + 2B = 0 \longrightarrow \frac{9}{2} + 2B = 0 \Rightarrow B = -\frac{9}{4} \end{cases}$$

So $\boxed{y_p = \frac{3}{2}t - \frac{9}{4}}$

More generally, if our DE is

$$ay'' + by' + cy = Ct^m$$

then our guess for a solution may be

$$y_p = A_m t^m + \cdots + A_1 t + A_0$$

t^5

$A_5 t^5 + A_4 t^4 + A_3 t^3 + A_2 t^2 + A_1 t + A_0$

$y = C \Rightarrow C = ?$

2. $y'' + 3y' + 2y = 10e^{3t}$ ← exponential

Guess: $y_p = Ae^{3t}$, $y' = 3Ae^{3t}$, $y'' = 9Ae^{3t}$

So $y'' + 3y' + 2y = (9Ae^{3t}) + 3(3Ae^{3t}) + 2Ae^{3t} = 10e^{3t}$

$\Rightarrow \underbrace{20A}e^{3t} \overset{\text{want}}{=} \underbrace{10}e^{3t}$

$\Rightarrow 20A = 10 \Rightarrow A = \frac{1}{2}$

So $\boxed{y_p = \frac{1}{2}e^{3t}}$

More generally, if our DE is

$$ay'' + by' + cy = Ce^{\alpha t}$$

then our guess for a solution may be

$$y_p = Ae^{\alpha t}$$

3. $y'' + 3y' + 2y = \sin t$

Guess: $y_p = \cancel{A\sin(t)}$

$y' = A\cos(t)$, $y'' = -A\sin(t)$

$-A\sin(t) + 3A\cos(t) + 2A\sin(t) = \sin(t)$

$\Rightarrow A\sin(t) + 3A\cos(t) = \sin(t) + 0\cos(t)$

$\Rightarrow \begin{cases} A = 1 \\ 3A = 0 \end{cases}$ ✗

$y_p = A\sin(t) + B\cos(t)$

$y_p' = A\cos(t) - B\sin(t)$

$y_p'' = -A\sin(t) - B\cos(t)$

$y'' + 3y' + 2y = (-A\sin(t) - B\cos(t)) + 3(A\cos(t) - B\sin(t)) + 2(A\sin(t) + B\cos(t)) \underset{\text{want}}{=} \sin(t)$

$\Rightarrow \sin(t)(-A - 3B + 2A) + \cos(t)(-B + 3A + 2B) = \sin(t)$

$\Rightarrow (A - 3B)\sin(t) + (B + 3A)\cos(t) = 1\cdot\sin(t) + 0\cdot\cos(t)$

$\Rightarrow \begin{cases} A - 3B = 1 & (1) \\ 3A + B = 0 & (2) \end{cases}$

$3(1) - (2)$: $-10B = 3 \Rightarrow \boxed{B = -\frac{3}{10}}$

$\Rightarrow 3A = \frac{3}{10} \Rightarrow \boxed{A = \frac{1}{10}}$

so $\boxed{y_p = \frac{1}{10}\sin(t) - \frac{3}{10}\cos(t)}$

More generally, if our DE is

$$ay'' + by' + cy = C\sin(\beta t) \quad \text{or } (C\cos(\beta t))$$

then our suggested solution is

$$y_p = A\cos(\beta t) + B\sin(\beta t)$$

poly · exp

$(A_2t^2 + A_1t + A_0)\cdot \cancel{A}e^t$

4. $y'' + 4y = 5t^2e^t$

Guess: $y_p = (A_2t^2 + A_1t + A_0)e^t$

$$y_p' = (A_2t^2 + A_1t + A_0)e^t + (2A_2t + A_1)e^t$$

$$y_p'' = [(A_2t^2 + A_1t + A_0)e^t + (2A_2t + A_1)e^t] + [(2A_2t + A_1)e^t + (2A_2)e^t]$$

$$y'' + 4y = [(A_2t^2 + A_1t + A_0)e^t + (2A_2t + A_1)e^t] + [(2A_2t + A_1)e^t + (2A_2)e^t] + 4((A_2t^2 + A_1t + A_0)e^t) = 5t^2e^t$$

$$\Rightarrow e^t[5A_2t^2 + (A_1 + 2A_2 + 2A_2 + 4A_1)t + (A_0 + A_1 + A_1 + 2A_2 + 4A_0)] = 5t^2e^t$$

$$\Rightarrow \begin{cases} 5A_2 = 5 \longrightarrow A_2 = 1 \\ 5A_1 + 4A_2 = 0 \longrightarrow 5A_1 = -4 \Rightarrow A_1 = -\frac{4}{5} \\ 5A_0 + 2A_1 + 2A_2 = 0 \rightarrow 5A_0 - \frac{8}{5} + 2 = 0 \Rightarrow 5A_0 = -\frac{2}{5} \end{cases}$$

$$\Rightarrow A_0 = -\frac{2}{25}$$

More generally, if our DE is

$$ay'' + by' + cy = Ct^me^{\alpha t}$$

so $\boxed{y_p = \left(t^2 - \frac{4}{5}t - \frac{2}{25}\right)e^t}$

then our suggested solution is

$$y_p = (A_mt^m + \cdots + A_0)e^{\alpha t}$$

Unfortunately, this method of choosing solutions doesn't always work.

5. $y'' + y' = 5$ $\quad y'' + y' = 0 + 0 \neq 5$???

Guess: $y_p = A$, $y_p' = 0$, $y_p'' = 0$

Look at $y'' + y' = 0$ char eqn: $r^2 + r = 0$

$r(r+1) = 0$

$r = 0, -1$

$y_h = c_1 + c_2 e^{-t}$

homogeneous

5 is in the general solution to the associated homogeneous equation.

The problem exists because our suggested function $y_p = A$ is also a solution to the homogeneous equation

$$y'' + y' = 0.$$

If $(r-1)^2 = 0$ as char eqn: $r = 1, 1$ $c_1 e^t + c_2 t e^t$

guess: $y_p = A \cdot t$ (not in the homogeneous solution)

$y_p' = A$

$y_p'' = 0$

$y'' + y' = 0 + A = 5 \Rightarrow A = 5$ ✓

Thus $\boxed{y_p = 5t}$

$y = Ae^{3t}$

6. $y'' - 6y' + 9y = e^{3t}$

$y'' - 6y' + 9y = 0$ char eqn: $r^2 - 6r + 9 = 0$

$(r-3)^2 = 0$

$r = 3, 3$

$y_h = c_1 e^{3t} + c_2 t e^{3t}$

Our suggested solution $y_p = At^2 e^{3t}$

$= t^2[Ae^{3t}]$

$y_p = At^2 e^{3t}$, $y' = (2At)e^{3t} + 3At^2 e^{3t} = (3At^2 + 2At)e^{3t}$

$y'' = (6At + 2A)e^{3t} + (3At^2 + 2At)(3e^{3t})$

$= (9At^2 + 12At + 2A)e^{3t}$

$y'' - 6y' + 9y = (\cancel{9At^2} + \cancel{12At} + 2A)e^{3t} - 6(\cancel{3At^2} + \cancel{2At})e^{3t}$

$+ 9(\cancel{At^2 e^{3t}}) = e^{3t}$

$\Rightarrow 2Ae^{3t} = e^{3t} \Rightarrow A = \frac{1}{2}$

so $\boxed{y_p = \frac{1}{2}t^2 e^{3t}}$

Recall in section 4.2, when we have a repeated root to the characteristic equation of a homogeneous DE, we add an extra factor of t to the proposed solution. So here, for #5, try $y_p = At$.

Statement: If a particular solution y_p contains terms that duplicate terms in the solution of the associated homogeneous DEQ, then y_p must be multiplied by t^s where s is the **smallest positive integer that eliminates that duplication**.

Case 1: No function in the assumed particular solution is a solution of the associated homogeneous DE. (See examples 1–4)

Case 2: A function in the assumed particular solution is also a solution of the associated homogeneous DE. (See examples 5–6)

Now try examples 5 and 6 again:

ex. Find a particular solution to $y'' + y' = 5$.

See previous

ex. Find a particular solution to $y'' - 6y' + 9y = e^{3t}$

See previous

Method of Undetermined Coefficients:

To find a particular solution to the DE

$$ay'' + by' + cy = f(t),$$

use the form $y_p(t)$ in the following tables.

$f(t)$	$y_p(t)$
$P(t)$	$t^s Q(t)$, deg $Q =$ deg P
$e^{\alpha t}$	$At^s e^{\alpha t}$
$e^{\alpha t}P(t)$	$t^s e^{\alpha t} Q(t)$, deg $Q =$ deg P

polynomial

$s = 0$ if α is not a root of the char. eqn.
$s = 1$ if α is a single root of the char. eqn.
$s = 2$ if α is a repeated root of the char. eqn.

$f(t)$	$y_p(t)$
$e^{\alpha t}\begin{cases}\cos\beta t\\ \sin\beta t\end{cases}$	$t^s e^{\alpha t}(A\cos\beta t + B\sin\beta t)$
$P(t)e^{\alpha t}\begin{cases}\cos\beta t\\ \sin\beta t\end{cases}$	$t^s e^{\alpha t}[Q_1(t)\cos\beta t + Q_2(t)\sin\beta t)]$

$s = 0$ if $\alpha + i\beta$ is not a root of the char.eqn.
$s = 1$ if $\alpha + i\beta$ is a root of the char. eqn.

ex. Find the form of a particular solution to the given DE:

1. $y'' + 2y' - 3y = f(t)$

homogeneous $y'' + 2y' - 3y = 0 \Rightarrow$ char eqn: $r^2 + 2r - 3 = 0$

$(r+3)(r-1) = 0$

$r = 1, -3$

$f(t)$	$y_p(t)$
$7\cos 3t$	$A\cos(3t) + B\sin(3t)$
$\alpha + i\beta$ $\alpha = 0$, $\beta = 3$, $3i$ ✗	
$2te^t \sin t$	$e^t\left[(A_1 t + A_0)\cos(t) + (B_1 t + B_0)\sin(t)\right]$
$r = 1 + i$	
$t^2 \cos(\pi t)$	$(A_2 t^2 + A_1 t + A_0)\cos(\pi t) + (B_2 t^2 + B_1 t + B_0)\sin(\pi t)$
$r = \pi i$	
$5e^{-3t}$	$t\left[Ae^{-3t}\right]$
$r = -3$ ✓	
$t^2 e^t$	$t\left[(A_2 t^2 + A_1 t + A_0)e^t\right]$
$r = 1$ ✓	

2. $y'' - 2y' + y = f(t) \longrightarrow y'' - 2y' + y = 0$ associated homogeneous equation

$\Rightarrow$ char eqn:

$r^2 - 2r + 1 = 0$

$\Rightarrow (r-1)^2 = 0$

$r = \underline{\underline{1, 1}}$

$f(t)$	$y_p(t)$
$7\cos 3t$	$A\cos(3t) + B\sin(3t)$
$r = \pm 3i$ X	
$2te^t \sin t$	$e^t\left[(A_1 t + A_0)\cos(t) + (B_1 t + B_0)\sin(t)\right]$
$r = 1 \pm i$	
$t^2 \cos(\pi t)$	$(A_2 t^2 + A_1 t + A_0)\cos(\pi t) + (B_2 t^2 + B_1 t + B_0)\sin(\pi t)$
$r = \pm \pi i$	
$5e^{-3t}$	Ae^{-3t}
$r = -3$	
$t^2 e^t$	$t^2\left[(A_2 t^2 + A_1 t + A_0)e^t\right]$
$r = 1$ ✓	number of repeats

3. $y'' - 2y' + 2y = 5te^t \cos t$

(Just find the form of a particular solution to the DE.)

$y'' - 2y' + 2y = 0 \Rightarrow r^2 - 2r + 2 = 0$

$$r = \frac{2 \pm \sqrt{4 - 4(1)(2)}}{2} = \frac{2 \pm \sqrt{-4}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

Next, $f(t) = 5t\underbrace{e^t \cos(t)}_{r = 1 \pm i}$ ✓

Use $y_p = t\left[e^t\left[(A_1 t + A_0)\cos(t) + (B_1 t + B_0)\sin(t)\right]\right]$

repeat

NOTE: The method of undetermined coefficients is **not** limited to second-order equations. It is limited to non-homogeneous linear equations $\cdots$

- that have constant coefficients, and
- where $f(t)$ is one of the 4 types: a polynomial function, an exponential function, a sine function, a cosine function, or finite sums and products of these functions.

ex. Determine whether or not the method of undetermined coefficients can be applied to find a particular solution of the given equation. $P(t)$, $e^{\alpha t}$, $\sin(\beta t)$, $\cos(\beta t)$

1. $y'' - y = \dfrac{e^t}{t^2} = t^{-2} e^t$ no

   rational

2. $y''' + 2y'' + y' = \dfrac{\cos 2t}{e^t} = e^{-t}\cos(2t)$ yes

3. $y'' + 4y = 2^t \sin 3t = (e^{\ln(2)})^t \sin(3t) = e^{\ln(2)t}\sin(3t)$

   yes

4. $y'' + ty' + t^2 y = \sin t$

   not constant coefficient eqn no

5. $y'' + y' - y = \tan t$ no

ex. Find a particular solution for $y'' - y = t\sin t$.

$y'' - y = 0 \Rightarrow r^2 - 1 = 0 \Rightarrow r = \pm 1$

$f(t) = t\sin(t) \longrightarrow r = \pm i \neq$

use $\quad y_p = (A_1 t + A_0)\cos(t) + (B_1 t + B_0)\sin(t)$

$$y_p' = [(A_1)\cos(t) + (A_1 t + A_0)(-\sin(t))] + [(B_1)\sin(t) + (B_1 t + B_0)\cos(t)]$$

$$y_p'' = -A_1\sin(t) + [(A_1)(-\sin(t)) + (A_1 t + A_0)(-\cos(t))] + B_1\cos(t) + [(B_1)\cos(t) + (B_1 t + B_0)(-\sin(t))]$$

$$= (-A_1 t - A_0 + 2B_1)\cos(t) + (-B_1 t - B_0 - 2A_1)\sin(t)$$

$$y'' - y = [(-A_1 t - A_0 + 2B_1)\cos(t) + (-B_1 t - B_0 - 2A_1)\sin(t)] - [(A_1 t + A_0)\cos(t) + (B_1 t + B_0)\sin(t)] = t\sin(t)$$

$$\Rightarrow \begin{cases} -2B_1 = 1 & t\sin t \longrightarrow B_1 = -\frac{1}{2} \\ -2A_1 = 0 & t\cos t \longrightarrow A_1 = 0 \\ -2A_0 + 2B_1 = 0 & \cos t \longrightarrow -2A_0 = 1 \Rightarrow A_0 = -\frac{1}{2} \\ -2A_1 - 2B_0 = 0 & \sin t \longrightarrow B_0 = 0 \end{cases}$$

$$\boxed{y_p = -\frac{1}{2}\cos(t) - \frac{1}{2}t\sin(t)}$$

How a forcing function changes the solution.

Consider the two IVPs:

(1) $y'' + y = 0$, $y(0) = 0$, $y'(0) = 1$.

(2) $y'' + y = \sin(2t)$, $y(0) = 0$, $y'(0) = 1$.

Solution for (1):

(1) $y'' + y = 0$, $y(0) = 0$, $y'(0) = 1$.

Characteristic equation: $r^2 + 1 = 0$; yields the solution $y_h(t) = c_1 \cos t + c_2 \sin t$

Apply initial conditions:

$y_h(0) = c_1 = 0$ then $c_1 = 0$

$y_h(t) = c_2 \sin t$

$y_h'(t) = c_2 \cos t$

$y_h'(0) = c_2 = 1$

then $y_h(t) = \sin t$

Solution for (2):

(2) $y'' + y = \sin(2t)$, $y(0) = 0$, $y'(0) = 1$.

$\alpha + i\beta = 2i$

$y_p(t) = t^s(A\cos(2t) + B\sin(2t))$

Characteristic equation: $r^2 + 1 = 0$; $r = \pm i$

$\alpha + i\beta = 2i$ is not a root of the characteristic equation, so $s = 0$

$y_p(t) = A\cos(2t) + B\sin(2t)$

$y_p'(t) = -2A\sin(2t) + 2B\cos(2t)$

$y_p''(t) = -4A\cos(2t) - 4B\sin(2t)$

insert into DE $y'' + y = \sin(2t)$ to solve for A and B

$-4A\cos(2t) - 4B\sin(2t) + A\cos(2t) + B\sin(2t) = \sin(2t)$

cos terms give $-3A = 0$ so $A = 0$

sin terms give $-3B = 1$ so $B = -\frac{1}{3}$

Then $y_p(t) = -\frac{1}{3}\sin(2t)$

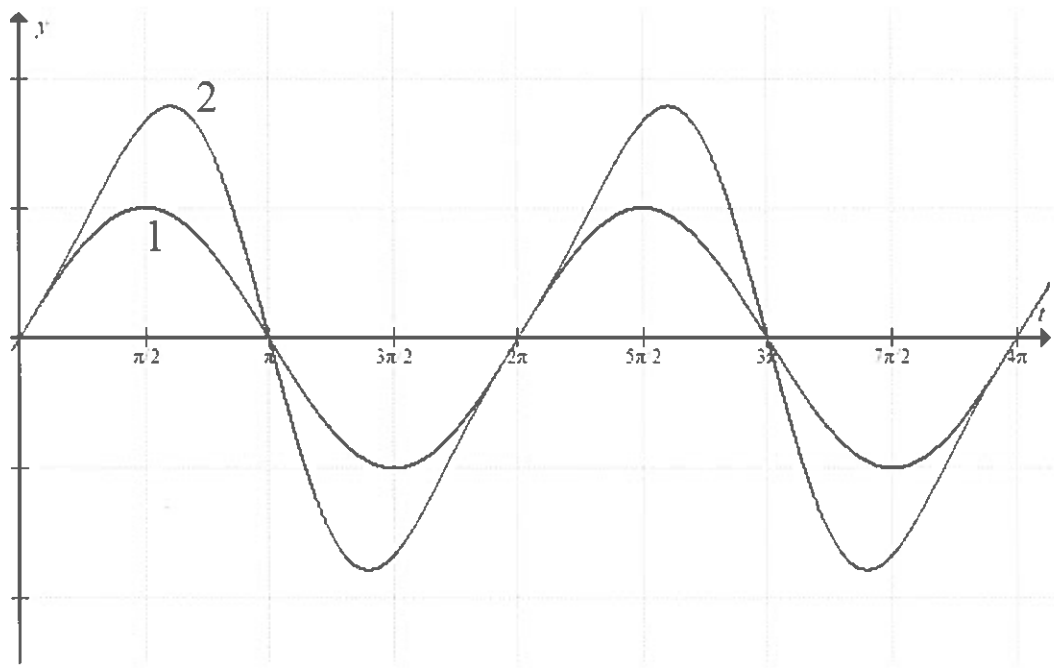
Now we have $y_g(t) = y_h(t) + y_p(t) = c_1 \cos t + c_2 \sin t - \frac{1}{3}\sin(2t)$.

Apply initial conditions to find c_1, c_2:

$$y_g(0) = c_1 = 0, \text{ then } c_1 = 0$$

$$y_g(t) = c_2 \sin t - \tfrac{1}{3}\sin(2t).$$

$$y_g'(t) = c_2 \cos t - \tfrac{2}{3}\cos(2t)$$

$$y_g'(0) = c_2 - \tfrac{2}{3} = 1, \qquad \text{then } c_2 = \tfrac{5}{3}.$$

$$y_g(t) = \tfrac{5}{3}\sin t - \tfrac{1}{3}\sin(2t)$$